

Data and Information Quality Project Report

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1 Introduction

The dataset provided for the project is a dataset from the Comune di Milano about personal service establishments. The raw file clearly shows typical data quality issues: missing values in some important fields, repeated records, and semi-structured text that mixes multiple pieces of information in a single cell. The goal of the project is therefore to implement data quality pipeline.

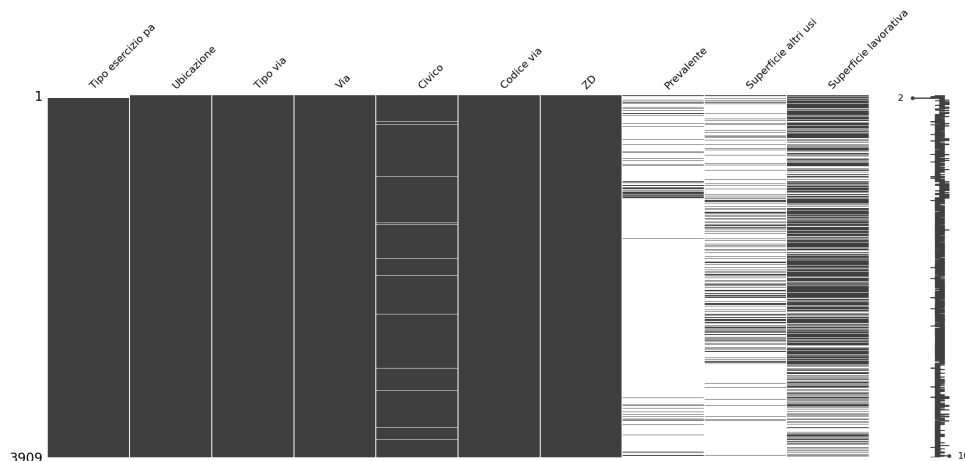
We first assessed the dataset with basic quality metrics, focusing on completeness and uniqueness and then we profiled the dataset. Based on these findings, we performed data wrangling to improve structure and consistency: we standardized text formats, transformed multi-value service descriptions into boolean indicators, and extracted address information from the composed *Ubicazione* field to populate structured columns. Finally, we applied specific rules to handle missing values and we performed duplicate detection to reduce redundancy.

2 Pipeline Implementation

2.1 Data Quality Assessment

The Data Quality Assessment focused on two primary dimensions: completeness and uniqueness.

The analysis revealed significant disparities in data completeness. While location-related columns, such as *Ubicazione* and *Via*, showed near-perfect coverage (approaching 100%), secondary fields like *Prevalente* and *Superficie altri usi* were highly sparse, with completeness rates of only 7.52% and 19.06% respectively. This contrast is visually summarized in the missing value matrix below, where the density of dark bands corresponds to the presence of data.



The uniqueness analysis provided insight into the nature of the dataset's attributes. For example extremely low uniqueness scores observed in columns such as *Tipo via* (0.43%) and *ZD* (0.25%) confirm their role as categorical variables with a limited cardinality.

These values function as valid sets for a multiple choice selection rather than free-text entry like *Ubicazione*. A high degree of repetition is consistent with the column meaning within these specific fields.

2.2 Data Profiling

The analysis of distinctness and uniqueness confirmed the findings from previous sections regarding the categorical nature of most variables. *Ubicazione* exhibits the highest degree of uniqueness (approximately 91%), which is expected as it functions as a composite field aggregating multiple pieces of information. In contrast, *Tipo esercizio pa* displays low uniqueness, consistent with its role as broad categorical classifier for commercial activities. While *Prevalente* shows some degree of distinctness, its utility is significantly limited by extreme data sparsity (only 7.5% completeness).

Regarding quantitative attributes, we analyzed the statistics for *Superficie lavorativa* and *Superficie altri usi*. The surface area values display a distribution consistent with the domain; however, the analysis suggests that potential outliers warrant further investigation. It is important to note that *Codice via*, despite being numeric, functions effectively as a categorical identifier.

Lastly, the association analysis revealed logical dependencies rather than unexpected correlations. Specifically, a strict link exists between *Codice via* and *Via*. Furthermore, we observed a naturally high association between *Via* and *ZD*, reflecting the inherent spatial hierarchy where streets are geographically nested within specific administrative zones.

2.3 Data Wrangling

The Data Wrangling phase started with column renaming to enhance readability (e.g., transforming *ZD* into *Municipio*) and dataset sorting. Subsequently, textual data was standardized by converting it to uppercase.

2.3.1 Column Tipo Esercizio

The *Tipo esercizio* column (originally labeled *Tipo esercizio pa*), which contained multiple concatenated values, was processed by transforming it into specific boolean variables (e.g. *ACCONCIATORE*, *ESTETISTA*) to uniquely classify the activities. This transformation was necessitated because many cells contained strings concatenated by semicolons, indicating that a single establishment could simultaneously offer multiple services. To address this issue, the following process was implemented:

- **Splitting:** The original column was parsed into multiple temporary columns by splitting the strings based on the semicolon delimiter.
- **Categorization:** Logical groups of keywords were defined to identify service macro-categories (e.g., the *Acconciatori* set includes terms such as 'PARRUCCHIERE PER SIGNORA', 'BARBIERE', and 'ACCONCIATORE').
- **Creation of Boolean Variables:** Rather than retaining a single, unstructured text column, five new boolean columns (True/False) were generated:
 - ACCONCIATORE
 - ESTETISTA
 - CENTRO ABBRONZATURA
 - TRATTAMENTO
 - TATUAGGI E PIERCING

This transformation converted the original multi-value field into a structured representation that is easier to analyze. Consequently, if an establishment operates as both a hairdresser and a beauty salon, it is simply marked as **True** in both respective columns, significantly facilitating data filtering and aggregation by service type.

To standardize the *Attività primaria* field, we derived it directly from the boolean service indicators created from *Tipo esercizio*. For each record, we checked whether at least one of the five service columns was marked as **True**. When this was the case, we assigned *Attività primaria* to the name of the first service column that was **True** (according to the fixed column order in the dataset). If none of the boolean columns was **True**, *Attività primaria* was left as missing.

2.3.2 Column Ubicazione

The transformation of the *Ubicazione* column was pivotal for retrieving and correcting missing data within the structured columns (*Via*, *Civico*, *Municipio*). This variable was selected as the primary reference specifically due to its superior data quality, characterized by high completeness and distinctness. This operation leveraged the fact that *Ubicazione* contained the complete address in a semi-structured string format (e.g., 'VIA CORREGGIO N. 8 (z.d. 7)'). The process was executed in two primary phases:

- **Regex Parsing:** A Regular Expression (Regex) was employed to deconstruct the location string and capture its individual constituents. The string analysis targeted specific patterns:
 - The first word for the *Street Type* (e.g., 'VIA', 'PZA').
 - The central segment for the *Street Name*.
 - The pattern 'N.' followed by digits to extract the *Civico*.
 - The pattern '(z.d. ...)' to extract the *Municipio*.
- **Imputation and Correction:** The extracted information was then used to update the structured columns. If a value was missing (e.g., *Civico* was null), we filled it using the value parsed from *Ubicazione*. Moreover, since we treated *Ubicazione* as the most reliable source for the full address, when an extracted value was available we also overwrote existing entries, even if they were not missing, in order to enforce consistency with the information contained in *Ubicazione*.

Once all valid information was transferred to *Tipo Via*, *Via*, *Civico*, and *Municipio*, the *Ubicazione* column, having become redundant, was removed from the dataset.

2.4 Null Values Handling

Missing values were handled with rules guided by the meaning of each column. The main idea was to avoid arbitrary filling, while still trying to avoid dropping rows where the most important columns weren't missing.

A first issue involved the activity description. Considering the transformations we did before on these columns, there is no possibility that some rows has just one of the columns values missing. Instead there is the possibility that some rows had both *Tipo esercizio* and *Attività primaria* missing: in this case, since the type of establishment could not be identified in any way and therefore, they were removed.

For the quantitative fields, we used simple imputations. *Superficie lavorativa* was filled using the median within a group defined by service type, which limits the impact of extreme values and keeps the filled values consistent with the activity. *Superficie altri usi* had a very large fraction of missing values; we decided to treat it as zero when not provided.

Finally, we filled missing values in the *Civico* column with 0. This choice was made because dropping rows where only the house number is missing would remove otherwise valid records. Using 0 allows us to keep these establishments in the dataset while clearly marking that the civic number is unknown.

For the remaining columns, a meaningful imputation was not feasible: missing values had no reliable information elsewhere in the dataset to infer them from, so any fill would have been essentially arbitrary. For this reason we removed the remaining rows containing nulls.

2.5 Outlier Detection

Outlier detection was considered during the project planning, but it was not applied in the final pipeline. The dataset contains only two numerical columns (*Superficie lavorativa* and *Superficie altri usi*) and they are not the main drivers of the analysis. Moreover, unusually large or small surface values can be valid in this domain, so removing them based on generic statistical rules risked deleting correct records without a clear benefit.

2.6 Duplicate Detection

We first removed exact duplicates, i.e., records that were identical across the relevant fields. After that, we addressed near-duplicates, which may differ only because of small spelling differences, formatting inconsistencies, or partial information. To make the process efficient, we used the *Sorted Neighbourhood* method: records were sorted using a key built from identifying information (*Tipo via* and *Via*), and comparisons were performed only within a small moving window.

Candidate pairs were then evaluated by comparing the main identifying fields of an establishment. For fields that are expected to be identical (*Tipo via*, *Municipio*, *Codice via*) we used exact matching. For fields that may contain small variations (typos, different formatting), we used Jaro–Winkler string similarity on *Via*, *Civico*, and *Attività primaria*.

The comparison outputs were combined into a single score by summing the feature values, and we classified a pair as a duplicate only when the score was very close to the maximum (threshold $T = 5.5$ out of 6).

When a pair was classified as a duplicate, we kept one record and removed the other. Before dropping the redundant row, we merged the five service boolean columns with a logical OR, so that if the two records contained different service flags, the final kept record preserves the union of the information.

3 Results and Conclusion

Overall, the pipeline produced a cleaner and more structured version of the dataset.

The main improvements produced by the pipeline can be summarized in these points:

- The multi-value *Tipo esercizio* field was transformed into boolean service indicators.
- Address-related fields were standardized and completed by parsing *Ubicazione* and transferring its content into *Tipo Via*, *Via*, *Civico*, and *Municipio*.
- The dataset was then processed to address missing values using simple imputations.
- Duplicate records were reduced through exact-duplicate removal and similarity-based matching.

After these steps, the dataset contains no nulls, fewer duplicated entries, and a more consistent structure compared to the raw input.